

MOHAMMED BAHEY EL-DEEN

Full Stack Software Engineer | Backend Specialist

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PROFESSIONAL SUMMARY

Backend-first full stack engineer specializing in Node.js/TypeScript microservices, distributed systems, and event-driven architecture. Production experience across fintech, AI compliance, and developer tooling — with React through end-to-end project delivery. Focused on clean, testable backend systems built to scale.

WORK EXPERIENCE

Software Developer — Reglint (AI Compliance SaaS · Arlington, VA — Remote) | Apr – May 2026

Python · AWS Lambda · Web Scraping · AWS S3 · Bedrock RAG

- Developed Python AWS Lambda functions scraping regulatory texts from 50 US state law portals (BIPA, CDPA, CPA, CTDP), feeding the knowledge corpus that powers Reglint's AI regulation-analysis agents via AWS Bedrock RAG.
- Contributed to test coverage for the AI compliance agents, validating BLOCK, REDACT, ALERT, and PASS enforcement paths across OFAC sanctions (fintech), HIPAA (healthcare), and GDPR/CCPA (privacy) violation scenarios.

Backend Developer Intern — Madkhol (Saudi Fintech · Remote) | Feb – May 2026

TypeScript · Node.js · Redis · Decimal.js · Microservices · TypeORM

- Traced reconciliation drift across 5 microservices to an FX re-derivation bug; engineered a provider-aware rounding framework standardized across the shared core package.
- Designed a Redis TTL-driven subscription state machine (ACTIVE → GRACE_ACTIVE → EXPIRED) closing two security gaps that allowed expired users to invest; backed by unit tests.
- Reviewed a force-liquidation cron PR, flagging migration dependencies, race conditions, and CMA regulatory compliance risks around selling user holdings.

PROJECTS

Auto-Investment Engine github.com/Hamshary99/Auto-Investment-Engine

TypeScript · Node.js · PostgreSQL · RabbitMQ · Docker Compose

- Engineered a 4-service fintech microservices system (API gateway, auth, portfolio manager, cron scheduler) with HMAC-SHA256 signed inter-service tokens and idempotent order execution via a PostgreSQL inbox table.
- Structured cron workers for daily NAV snapshots using Decimal.js precision math alongside a reconciliation loop that auto-fails pending orders past SLA.

Movies Reservation App github.com/Hamshary99/Movies-Reservation-App

Node.js · Express · React · MongoDB · JWT · Stripe

- Architected a full-stack cinema booking platform with a three-role RBAC system, 20+ CRUD endpoints, Stripe payment integration, and QR-code ticket scanning.
- Built the React frontend managing multi-step booking flows, seat selection state, and payment confirmation UI.

AI Assistant for Developers github.com/Hamshary99/AI-Assistant-for-Developers

Node.js · Express · React · MongoDB · Docker · Gemini API · Zod

- Built 6 AI-powered endpoints (explain, fix, compare, README, suggestions, history) integrated with Gemini API; enforced via Zod validation schemas and global rate limiting.
- Developed a React frontend with live Markdown rendering, syntax highlighting, and code diffs; containerized via Docker Compose multi-stage pipelines.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C/C++

Backend: Node.js, NestJS, Express.js, RESTful APIs, Microservices, RabbitMQ, Cron Services

Frontend: React.js (hooks, component lifecycle, controlled state)

Databases: PostgreSQL, MongoDB, MySQL, Redis, TypeORM

Cloud & DevOps: AWS Lambda, AWS S3, Docker, GitHub Actions

Foundations: OOP, Data Structures & Algorithms, Clean Architecture

EDUCATION

B.S. Mechatronics Engineering — Menoufia University

Languages: Arabic — Native · English — Fluent